# LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

## Quick Revision: Force and Pressure

### Key ideas

- A **force** is a push or a pull. It arises from an interaction between two objects, and has both size (magnitude) and direction. SI unit: newton (N).
- A force can **start or stop motion, change speed, change direction, or change shape**.
- Forces in the same direction add up; forces in opposite directions subtract. Equal and opposite forces cancel.

| Contact forces | Non-contact forces |
|---|---|
| Muscular force (our muscles), friction (between surfaces in contact) | Magnetic force, electrostatic force (charged objects), gravitational force (Earth's pull) |

**Pressure = force ÷ area** · unit: $N/m^2$ = pascal (Pa)
Same force, smaller area → more pressure. Same force, larger area → less pressure.

### Pressure in daily life

- Knives, nails and needles have sharp ends (small area, high pressure). School bags have wide straps and porters put a round cloth on their heads (large area, low pressure).
- **Liquids** press on the bottom and the walls of their container, and the pressure **increases with depth**. That is why dam walls are thicker at the bottom.
- **Air** around us exerts atmospheric pressure. A rubber sucker sticks to a smooth wall because the air pressure outside pushes it.

### Common mistakes

- Writing pressure without a unit, or mixing up force (N) and pressure (Pa).
- Saying a force always makes things move. It may only change their shape.

### Quick check

1. A force of 100 N acts on an area of 2 $m^2$. What is the pressure?

2. Why do heavy trucks and tractors have broad tyres?